

Pediatrics and child health

OSCE2

Endocrinology

Hypothyroidism

CONGENITALHYPOTHYROIDISM

- Most cases of congenital hypothyroidism are not hereditary and can occur sporadically and result from thyroid dysgenesis (anatomical).
- Some cases may be familial, usually caused by one of the inborn errors of thyroid hormone synthesis, and may be associated with a goiter (Functional).
- In many cases, the deficiency of thyroid hormone is severe, and symptoms develop in the early weeks of life.
- The prevalence of congenital hypothyroidism based on nationwide programs(USA) for neonatal screening is 1/4,000 infants worldwide
- Girls are more affected than boys(2X).

CONGENITAL HYPOTHYROIDISM

1.THYROID DYSGENESIS

2.DYSHORMONOGENESIS

3.Ingestion of goiterogenous substance by the mother during pregnancy

4.Iodine deficiency-

5.Hypothalamus and pituitary defect

6.TH unresponsiveness

Clinical Manifestation

- Asymptomatic at birth even in case of agenesis
- They have low serum T4 and elevated TSH and this will be detected during newborn screening.
- F:M 2:1, birth wt and length normal but head size may be slightly increased
- **Prolonged physiologic jaundice, delayed passage of stool**
- Affected infants cry little, sleep much, have poor appetites, and are generally sluggish.
- The abdomen is large, and an umbilical hernia is usually present.
- The temperature is subnormal
- Edema of the genitals and extremities may be present.

Clinical Manifestation





Investigation

- ✓ T3,T4 and TSH
- ✓ Delayed bone age in around 60% of the cases.
- ✓ Scintigraphy can help to pinpoint the underlying causes in infants with congenital hypothyroidism
- ✓ ECG usually shows low voltage P and T waves with diminished QRS complex due to poor LV function and pericardial effusion.

Treatment

L-thyroxine is the drug with a dose of 10 microgram/kg/day for neonates and 4 microgram/kg/day for children beyond neonatal age.

Prognosis

Early diagnosis and treatment from the first week of life will result in good prognosis.

Delay in diagnosis ,inadequate treatment and poor compliance will result in variable brain damage

Around 20 % of children will have a neuro sensorial hearing loss and without treatment affected infants become profoundly mentally deficit dwarfs.

Graves Disease

- Graves disease occurs in **0.02%** of children
- peak incidence in the **11- to 15-yr** of age
- There is a **5 : 1** female to male ratio.
- **Most have a positive family history**
- Enlargement of the thymus, splenomegaly, lymphadenopathy, infiltration of the thyroid gland and retro-orbital tissues with lymphocytes and plasma cells, and peripheral lymphocytosis are well-established findings in Graves disease

Clinical manifestation

- **5%** of all patients with hyperthyroidism are <15 yr of age
- The peak incidence occurs during adolescence
- Graves disease has begun **between 6 wk and 2 yr** of age in children born to mothers without a history of hyperthyroidism
- The clinical course in children is **highly variable** but usually is not so fulminant as in many adults

MANIFESTATIONS OF HYPERTHYROIDISM

• Symptoms

- Hyperactivity, irritability, altered mood, insomnia, anxiety
- Heat intolerance, increased sweating
- Palpitations Fatigue, weakness ,Dyspnea
- Weight loss with increased appetite, Pruritus , Increased stool frequency,
- thirst and polyuria, Oligomenorrhea or amenorrhea

• Signs

- Sinus tachycardia, atrial fibrillation, supraventricular tachycardia
- Fine tremor, hyperkinesia, hyperreflexia , Warm, moist skin
Palmar erythema, onycholysis ,Hair loss,
Osteoporosis
- Muscle weakness and wasting, High-output heart failure
- Chorea,Periodic (hypokalemic) paralysis ,Psychosis

MANIFESTATIONS OF GRAVES DISEASE

- Diffuse goiter
- Ophthalmopathy
 - A feeling of grittiness and discomfort in the eye,
Retrobulbar pain, Eyelid lag or retraction
Periorbital edema, chemosis, scleral injection
Exophthalmos, Extraocular muscle dysfunction
Exposure keratitis, Optic neuropathy
- Localized dermopathy (rare in children)
- Lymphoid hyperplasia

Conditions associated with graves

- Type 1 diabetes mellitus
- Addison disease
- Vitiligo
- Pernicious anemia
- Alopecia areata
- Myasthenia gravis
- Celiac disease

Thyroid crisis or thyroid storm

- **is** a form of hyperthyroidism manifested by an acute onset, hyperthermia, severe tachycardia, heart failure, and restlessness
- May be rapid progression to delirium, coma, and death
- Precipitating events include trauma, infection, radioactive iodine , or surgery
- These symptom complexes are rare in children

Laboratory findings

- T_4 and T_3 are elevated
- Levels of TSH are suppressed
- Antithyroid antibodies are often present
- Radioiodine is rapidly and diffusely concentrated in the thyroid
- Bone density may be reduced at diagnosis but returns to normal with treatment

Treatment

- Initial medical therapy using antithyroid drugs rather than radioiodine or subtotal thyroidectomy
- Radioiodine is gaining acceptance as initial treatment in children **>10 yr of age**
- All therapeutic options have advantages and disadvantages
- The 2 antithyroid drugs used historically are **propylthiouracil (PTU)** and **methimazole**
- Both compounds inhibit incorporation of trapped inorganic iodide into organic compounds, and they might also suppress TRSAb levels by directly affecting intrathyroidal autoimmunity

Table 562-4 -- MANAGEMENT OF THYROID STORM IN ADOLESCENTS

GOAL	TREATMENT
Inhibition of thyroid hormone formation and secretion	Propylthiouracil (PTU), 400 mg every 8 hr PO or by nasogastric tube Sodium iodide, 1 g IV in 24 hr, or saturated solution of KI, 5 drops every 8 hr
Sympathetic blockade	Propranolol, 20-40 mg every 4-6 hr, or 1 mg IV slowly (repeat doses until heart rate slows); not indicated in patients with asthma or heart failure that is not rate related
Glucocorticoid therapy	Hydrocortisone, 50-100 mg IV every 6 hr
Supportive therapy	Intravenous fluids (depending on indication: glucose, electrolytes, multivitamins) Temperature control (cooling blankets, acetaminophen; avoid salicylates) O ₂ if required Digitalis for heart failure and to slow ventricular response; pentobarbital for sedation Treatment of precipitating event (e.g., infection)

Congenital Hyperthyroidism

- Neonatal Graves disease is caused by **transplacental passage of TRSAb**
- If the infant has been exposed to antithyroid drugs, onset of symptoms is **delayed by 3-4 days**
- If TRBAb is also present, onset of hyperthyroid symptoms may be **delayed for several weeks**
- **Neonatal hyperthyroidism** occurs in only **2%** of infants born to mothers with **Graves disease**
- Male and female affects equally
- The disorder usually **remits spontaneously within 6-12 wk** but can persist longer, depending on the levels of TRSAb
- Mild asymptomatic hyperthyroxinemia also occurs
- Have family histories of Graves disease

Clinical Manifestations

- Most are premature and, have IUGR and goiters
- The infant is extremely restless, irritable, and hyperactive and appears anxious and unusually alert
- Microcephaly and ventricular enlargement
- Eyes are opened widely and exophthalmic
- Extreme tachycardia and tachypnea, and elevated To.
- In severely affected infants, there is a progression of symptoms; weight loss occurs despite a ravenous appetite, hepatosplenomegaly, and jaundice
- Severe hypertension and cardiac failure
- Can die if therapy is not instituted promptly
- T_4 and T_3 are markedly elevated, and TSH is suppressed
- Advanced bone age, frontal bossing with triangular facies, and cranial synostosis are common, especially in infants with persistent clinical manifestations of hyperthyroidism

Treatment

- Oral administration of **propranolol** (1-2 mg/kg/24 hr, orally in 3 divided doses) and **methimazole** (0.25-1.0 mg/kg/24 hr given every 12 hr); **Lugol solution** (1 drop every 8 hr) may be added.
- If the thyrotoxic state is severe, **parenteral fluid** therapy and corticosteroids
- After a euthyroid state is reached, only antithyroid drug treatment is necessary
- The dose should be gradually tapered to keep the infant euthyroid
- Most cases remit by 3-4 mo of age

Prognosis

- Advanced osseous maturation, microcephaly, and mental retardation occur with delayed treatment
- Intellectual development is normal in most treated infants
- **In utero hyperthyroidism appears to suppress the hypothalamic-pituitary-thyroid feedback mechanism, and develop permanent central hypothyroidism, requiring lifelong thyroid hormone treatment**

DKA management

General Resuscitation Measure

- Acute management should follow the general guidelines for PALS
 - A→If comatose insert NGT and maintain the airway
 - B→Give oxygen by face mask for patients with severe circulatory impairment or shock
 - C→Insert IV cannula and take blood sample and if the child is in shock give **20 ml/kg of 0.9 % normal** saline to the maximum of 30 ml/kg depending on the response of the child over 30 minutes
1. Expand intravascular volume
 2. Rehydration therapy
 3. Potassium replacement therapy
 4. Insulin therapy
 5. Avoid complications (Hypoglycemia, Cerebral edema and Hypokalemia)
 6. Treat precipitating factors
 7. Follow up

Fluid therapy

- **Expand intravascular volume**
- Bolus fluid 10 - 20 ml /kg 0.9 % N/S to be given over 1-2 hours can be repeated if necessary
 - ✓ should be gradual and with isotonic fluids or ringers lactate.
 - ✓ repeat the same volume if circulating volume is still compromised.
- Maintenance fluid /48 hours + Deficit fluid - Bolus fluid = -----ml/ 48 hr with potassium 40 mmol in 1000 ml of normal saline.
 - Deficit=85ml/kg

Potassium

- K + Replacement therapy is required regardless of the serum potassium concentration.
- 40 mmol in 1000 ml of normal saline.
- Make sure that the child is passing urine before initiation of KCL.
- Can be given as potassium phosphate or acetate or chloride with a rate of 0.5 mmol/kg/hr (20-80 mmol/l of fluid)
- Administration of potassium entirely as potassium chloride contributes to the risk of hyperchloremic metabolic acidosis, whereas administration entirely as potassium phosphate can result in hypocalcemia
- Keep mannitol at the bed side (dose 0.5-1 gm/kg IV over 20 min)

INSULIN

- Start insulin infusion 1-2 hours after starting fluid management (Intravascular volume expansion)
- Regular insulin **0.1 u/kg/hour**.
- Low dose iv insulin administration should be the standard of care
- If continuous IV administration of insulin is not possible regular insulin SC or IM with a dose: 0.1 unit/kg every 1-2 hours can be given.
- If this is not practical
 - give 0.5 u/kg every 6 hourly (½ IM and ½ sc for the first dose)
 - continue with 0.5 u/kg sc every 4-6 hours.

cont...

- **Don't omit insulin is the key**
- If there is a rapid decline of glucose > 90 mg/dl /hour
 - decrease the dose to 0.05 u/kg/hr and
 - Change the fluid to ½ N/S in 10 % D/W.
 - Consider adding glucose even before plasma glucose has decreased to 250 mg - 300 mg/dl.
- Continue the management until the child is out of DKA (PH > 7.3 or bicarbonate >15 mmol/l)
- To prevent rebound hyperglycaemia the first SC injection should be given 1-2 hours (with Regular insulin) before stopping the insulin infusion.

cont...

- **Bicarbonate:** Indicated if there is only severe acidosis $\text{PH} < 6.9$
- **Oral fluids-** can be introduced when there is marked clinical improvement (mild acidosis or ketosis can be there)

Treat the infection

➤ Infection

➤ Stress

➤ Insulin omission

Complications of management

- Hypoglycemia
- Hypokalemia
- Cerebral edema

Management complications

Hypoglycemia

- Hypoglycemic Reactions (Insulin Shock): symptoms and signs include pallor, sweating, apprehension, trembling, tachycardia, hunger, drowsiness, mental confusion, seizures and coma.
- Insulin decrease the dose to 0.05 u/kg/hr and you can change the fluid to $\frac{1}{2}$ N/S in 10 % D/W. Consider adding glucose even before plasma glucose has decreased to 250 mg - 300 mg/dl.
- glucagon 0.5 mg is administered to an unconscious or vomiting child

Cerebral edema

- Major life-threatening complication seen in the treatment of children with DKA
 - Usually develops several hours after the institution of therapy
- Most commonly presents in
 - Children between 5 –14 years
 - With new onset diabetes
 - Longer duration of symptoms
 - Greater hypocapnia at presentation after adjusting for degree of acidosis
 - Increased serum urea nitrogen at presentation
 - More severe acidosis
 - Bicarbonate treatment for correction of acidosis

Cerebral edema

- Signs and symptoms of cerebral edema include:
 - Headache and slowing of heart rate
 - Change in neurological status (restlessness, irritability, increased drowsiness, and incontinence)
 - Specific neurological signs (e.g., cranial nerve palsies, papilledema)
 - Rising blood pressure
 - Decreased O2 saturation
- Diagnosis
 - Abnormal motor or verbal response to pain
 - Decorticate or decerebrate posture
 - Cranial nerve palsy (especially III, IV, and VI)
 - Abnormal neurogenic respiratory pattern

Cont.

Major criteria

- Altered mentation/fluctuating level of consciousness
- Sustained heart rate deceleration (decrease >20 beats/min) not attributable to improved intravascular volume or sleep state
- Age-inappropriate incontinence

Minor criteria

- Vomiting
- Headache
- Lethargy or not easily arousable
- Diastolic blood pressure >90mmHg
- Age <5 yr

One diagnostic, 2 major or one major and 2 minor criteria for diagnosis
92% sensitivity and 96% specificity

Treatment of CE

- Give mannitol **0.5-1 gram/kg IV over 20 minutes** and repeat if there is no initial response in **30 minutes to 2hrs**
- Reduce the rate of fluid administration by one third
- Hypertonic saline (3%), 5-10 mL/kg over 30 minutes, may be an alternative to mannitol, especially if there is no initial response to 1mg/kg mannitol.
- Mannitol should be available at Bedside
- Intubation may be necessary for the patient with impending respiratory failure, and hyperventilation (to a $pCO_2 < 22$ mm Hg [2.9 kPa])
- Elevate the head of the bed to 30°
- After treatment has been started, a CT scan should be obtained to rule out other possible intracerebral causes of neurologic deterioration

Monitoring

- We follow DKA patients with a DKA flow sheet which has the following components:
 - Date and Time
 - Vitals and RBS
 - Urine glucose
 - Urine ketone every 2 hour
 - Electrolyte(K + every (2-4 hrs)
 - Neurologic status (GCS) hourly
 - Insulin dose
 - Fluid input/output

Prevention

- Advise on adherence and appropriate storage of insulin(insulin omission commonest cause)
- Advise on what to do when they have poly symptom and other symptoms of DKA(stressful conditions and signs of an impending DKA)
- Advise on self monitoring of blood sugar

Adrenocortical Insufficiency

- In *primary adrenal insufficiency* it prevents production of *cortisol* and often *aldosterone*.
- Dysfunction of the *hypothalamus* or *anterior pituitary gland* may cause a deficiency of corticotropin (ACTH) and lead to hypofunction of the adrenal cortex; this is termed *secondary adrenal insufficiency*.

Etiologies

1. Congenital Adrenal Hyperplasia (Inborn defects of steroidogenesis).
2. Adrenal hypoplasia congenita
3. Adrenoleukodystrophy
4. Familial Glucocorticoid deficiency
5. Disorder of cholesterol synthesis

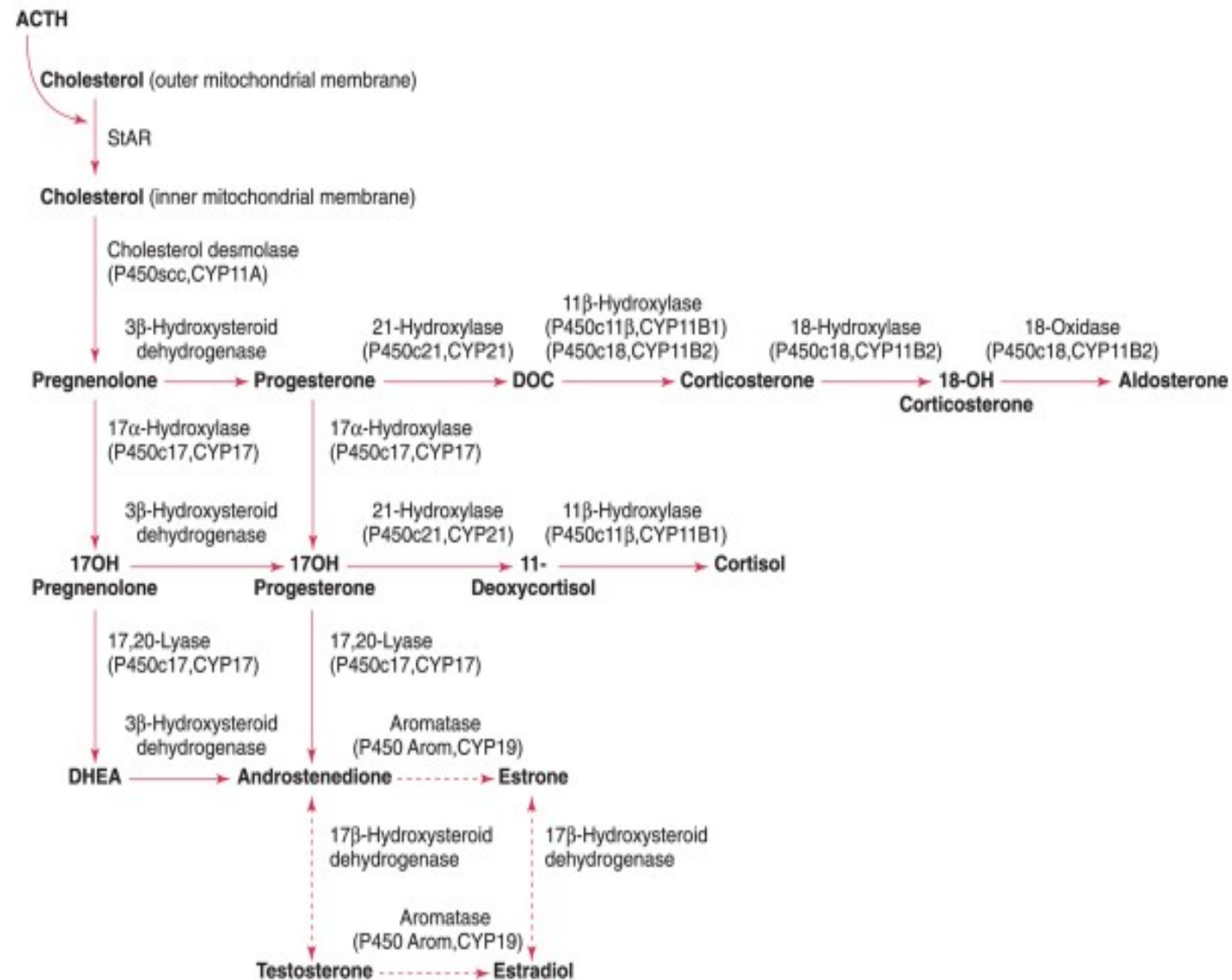
1. Congenital Adrenal Hyperplasia

- 75 % of Infants with 21-hydroxylase deficiency,
 - Almost all infants with lipoid adrenal hyperplasia, and
 - Most infants with a deficiency of 3 β -hydroxysteroid dehydrogenase
- Newborns with the above conditions will manifest with salt-losing symptoms because they are unable to synthesize **either cortisol or aldosterone**.

Depending on the enzymatic step that is deficient, there may be signs, symptoms, and laboratory findings of

- ❖ ***Incomplete virilization or premature puberty in affected males and***
- ❖ ***Virilization or sexual infantilism in affected females.***
- ❖ ***Mineralocorticoid deficiency or excess;***

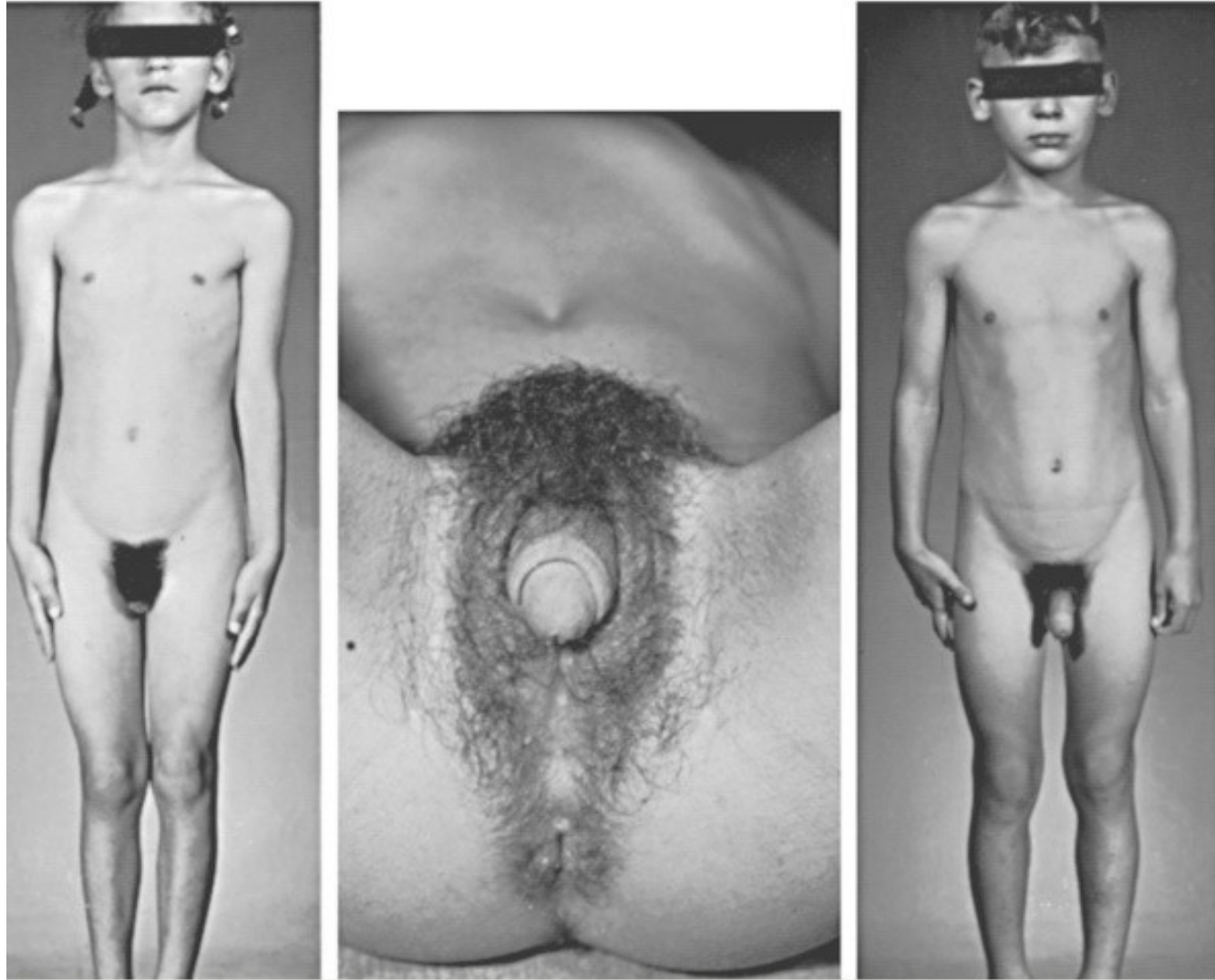
Steroidogenesis



Congenital Adrenal Hyperplasia

- 21 alpha hydroxylase deficiency is the commonest one and followed by 11 beta hydroxylase deficiency and 3B- dehydroxylase enzyme deficiency

Congenital Adrenal Hyperplasia



Laboratory findings

- ✓ Low Na⁺ and K⁺ is high,
- ✓ cortisol level is decreased and
- ✓ ACTH is increased in case of primary adrenal insufficiency.
- ✓ The most definitive test for adrenal insufficiency is measurement of serum levels of cortisol before and after administration of ACTH.

Treatment

- Treatment of acute primary adrenal insufficiency must be immediate and vigorous.
- Give 5 % D/W with 0.9 % N/S to correct hypoglycemia, hyponatremia and hypotension.
- If the hyperkalemia is severe it needs treatment. IV hydrocortisone should be given.
- After the acute manifestation is under control most patients require chronic replacement therapy for their cortisol and aldosterone deficiencies with a dose of 6-12 mg/m²/day hydrocortisone and 0.1 mg fludrocortisone replacement therapy.

Growth hormone



FIG. 576.2 Cerebral gigantism (Sotos syndrome) in an 8 yr old boy. The height age was 12 yr, and the bone age was 12 yr. IQ was 60. The electroencephalogram had abnormal findings. Note the prominence of the forehead and jaw and the large hands and feet. Sexual development was